L15

L15: Primitive Roots

The order of an integer.

- Let m be a positive integer and a be an integer relatively prime to m. Euler's Theorem tells us that $a^{\phi(m)} \equiv 1 \pmod{m}$. However, it may happen that some smaller positive power than $\phi(m)$, say n, will result in $a^n \equiv 1 \pmod{m}$.

Definition: Let $a, m \in \mathbb{Z}$ with $m > 0$ and $(a,m) = 1$. The order of a modulo m, denoted $\operatorname{ord}_m a$, is the least positive integer n for which $a^n \equiv 1 \pmod{m}$.

Example: • Consider $\operatorname{ord}_7 2$.

$2^1 \equiv 2 \pmod 7$, $2^2 \equiv 4 \pmod 7$, $2^3 \equiv 1 \pmod 7$.

Thus $\operatorname{ord}_7 2 = 3$.

Euler's Theorem guarantees that

$$\operatorname{ord}_7 2 \leq \phi(7) = 6.$$

- Consider $\text{ord}_7 3$.

$3^1 \equiv 3 \pmod 7$, $3^2 \equiv 2 \pmod 7$, $3^3 \equiv 6 \pmod 7$, $3^4 \equiv 4 \pmod 7$, $3^5 \equiv 5 \pmod 7$, $3^6 \equiv 1 \pmod 7$.

Thus $\text{ord}_7 3 = 6$. Note that Euler's Theorem guarantees that $\text{ord}_7 3 \leq \phi(7) = 6$.

- We have seen that Euler's Theorem guarantees that $\text{ord}_m a \leq \phi(m)$. The next Proposition gives us the stronger statement that $\text{ord}_m a \mid \phi(m)$.

<u>Proposition 5.1</u> Let $a, m \in \mathbb{Z}$ with $m > 0$ and $(a,m) = 1$. Then $a^n \equiv 1 \pmod m$ for some positive integer n if and only if $\text{ord}_m a \mid n$. In particular, $\text{ord}_m a \mid \phi(m)$.

<u>Proof</u>: ($\Rightarrow$ direction): Assume that $a^n \equiv 1 \pmod m$ for some positive integer n. By the Division algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$n = (\text{ord}_m a)q + r, \quad 0 \leq r < \text{ord}_m a.$$

Then $a^n = a^{(\mathrm{ord}_m a)q + r} = (a^{\mathrm{ord}_m a})^q a^r$ ③

$$\equiv a^r \pmod{m}.$$

Since $a^n \equiv 1 \pmod{m}$, we have that $a^r \equiv 1 \pmod{m}$. Now, $0 \leq r < \mathrm{ord}_m a$, and the definition of $\mathrm{ord}_m a$ as the least positive power of a that is congruent to 1 modulo m, together imply that $r = 0$. Thus $n = (\mathrm{ord}_m a)q$ and so $\mathrm{ord}_m a \mid n$, as desired.

($\Leftarrow$ direction): Assume that $\mathrm{ord}_m a \mid n$. Then there exists $b \in \mathbb{Z}$ with $b > 0$ such that $n = (\mathrm{ord}_m a) b$. Now,

$$a^n = a^{(\mathrm{ord}_m a) b} = (a^{\mathrm{ord}_m a})^b \equiv 1 \pmod{m},$$

as desired. ∎

Example: • By Proposition 5.1 we have that $\mathrm{ord}_7 2 \mid \phi(7) = 6$. Thus $\mathrm{ord}_7 2$ is 1, 2, 3 or 6. We showed previously that $\mathrm{ord}_7 2 = 3$

• By Proposition 5.1 we have that

$ord_{13} 2 \mid \phi(13) = 12$. Thus $ord_{13} 2$ is 1, 2, 3, 4, 6, or 12. We have:

$2^1 \equiv 2 \pmod{13}$, $2^2 \equiv 4 \pmod{13}$, $2^3 \equiv 8 \pmod{13}$, $2^4 \equiv 3 \pmod{13}$ and $2^6 \equiv 12 \pmod{13}$.

Since we know that $2^{12} \equiv 1 \pmod{13}$ by Euler's Theorom, we have that $ord_{13} 2 = 12$ (without having to compute $2^7, 2^8, 2^9, 2^{10}$, and $2^{11} \pmod{13}$).

<u>Proposition 5.2</u>: Let $a, m \in \mathbb{Z}$ with $m > 0$ and $(a,m) = 1$. If i and j are non-negative integers, then $a^i \equiv a^j \pmod{m}$ if and only if $i \equiv j \pmod{ord_m a}$.

<u>Proof</u>: Without loss of generality, we may assume that $i \geq j$.

($\Rightarrow$ direction): Assume that $a^i \equiv a^j \pmod{m}$, or, equivalently $a^j a^{i-j} \equiv a^j \pmod{m}$ (since $i \geq j$).

Since $(a^j, m) = 1$, the inverse of a^j, say $(a^j)'$, exists by Corollary 2.8. Multiplying both sides of the above congruence by $(a^j)'$, we obtain

$$(a^j)' a^j a^{i-j} \equiv (a^j)' a^j \pmod{m}.$$

Thus

$$a^{i-j} \equiv 1 \pmod{m}.$$

By Proposition 5.1, we have that $\text{ord}_m a \mid i-j$, from which $i \equiv j \pmod{\text{ord}_m a}$, as desired.

($\Leftarrow$ direction) Assume that $i \equiv j \pmod{\text{ord}_m a}$. Then $\text{ord}_m a \mid i-j$, and so there exists $n \in \mathbb{Z}$ with $n > 0$ such that $i-j = (\text{ord}_m a)n$. Now

$$\begin{aligned} a^i = a^{(\text{ord}_m a)n + j} &= a^{(\text{ord}_m a)n} a^j \\ &= (a^{\text{ord}_m a})^n a^j \\ &\equiv a^j \pmod{m}, \end{aligned}$$

as desired. ∎

<u>Example</u>: Let $a = 2$ and $m = 7$. Then

$ord_7 2 = 3$ by the previous example.

Proposition 5.2 implies that if i and j are non-negative integers, then $2^i \equiv 2^j \pmod 7$ if and only if $i \equiv j \pmod 3$. For example, since $2000 \equiv 2 \pmod 3$, we have that

$$2^{2000} \equiv 2^2 \equiv 4 \pmod 7.$$

<u>Definition</u>: Let $r, m \in \mathbb{Z}$ with $m > 0$ and $(r, m) = 1$. Then r is said to be a primitive root modulo m if $ord_m r = \phi(m)$.

<u>Example</u>: • 3 is a primitive root modulo 7 since $ord_7 3 = 6 = \phi(7)$ by a previous example.

• 2 is a primitive root modulo 13 since $ord_{13} 2 = 12 = \phi(13)$ by a previous example.

• 2 is not a primitive root modulo 7 since $ord_7 2 = 3 \neq \phi(7)$ by a previous example.

<u>Example</u>: Prove that there are no primitive

roots modulo 8. The reduced residues modulo 8 are 1, 3, 5 and 7. A straightforward check shows that $\mathrm{ord}_8 1 = 1$, $\mathrm{ord}_8 3 = 2$, $\mathrm{ord}_8 5 = 2$ and $\mathrm{ord}_8 7 = 2$. Since none of the above orders are equal to $\phi(8) = 4$, we have that there are no primitive roots modulo 8.

- Not all integers m possess primitive roots. The Primitive Root Theorem will determine precisely which integers have primitive roots (in a later lecture).

<u>Proposition 5.3</u>: Let r be a primitive root modulo m. Then

$$\{r, r^2, r^3, \ldots, r^{\phi(m)}\}$$

is a set of reduced residues modulo m.

<u>Remark</u>: This result says that a primitive root (if it exists), generates the set of reduced residue classes modulo m.

Proof: Since r is a primitive root modulo m, we have that $(r,m)=1$, and so each element of the set is relatively prime to m. Since there are $\phi(m)$ such elements, it remains to show that no two elements of the set are congruent modulo m. Assume that $r^i \equiv r^j \pmod{m}$ for $1 \le i,j \le \phi(m)$. Then Proposition 5.2 implies that $i \equiv j \pmod{\phi(m)}$, from which $i=j$ since $1 \le i,j \le \phi(m)$. This completes the proof. $\blacksquare$

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Example:

- 3 is a primitive root modulo 7 by a previous example. One can check

$$\{3, 3^2, 3^3, 3^4, 3^5, 3^6\}$$

$$= \{3,2,6,4,5,1\} \pmod{7}.$$

- 2 is a primitive root modulo 13 by a previous example. One can check that

$\{2, 2^3, \ldots, 2^{12}\}$ is the set of reduced residues modulo 13. One can check that this set is $\{1, 2, 3, \ldots, 12\} \pmod{13}$ (in some permuted order).

- If a primitive root modulo m exists, it usually is not unique modulo m. At the start of next lecture we will determine the number of primitive roots there are modulo m (provided at least one exists).

Exercise: Show there are no primitive roots modulo 12.